

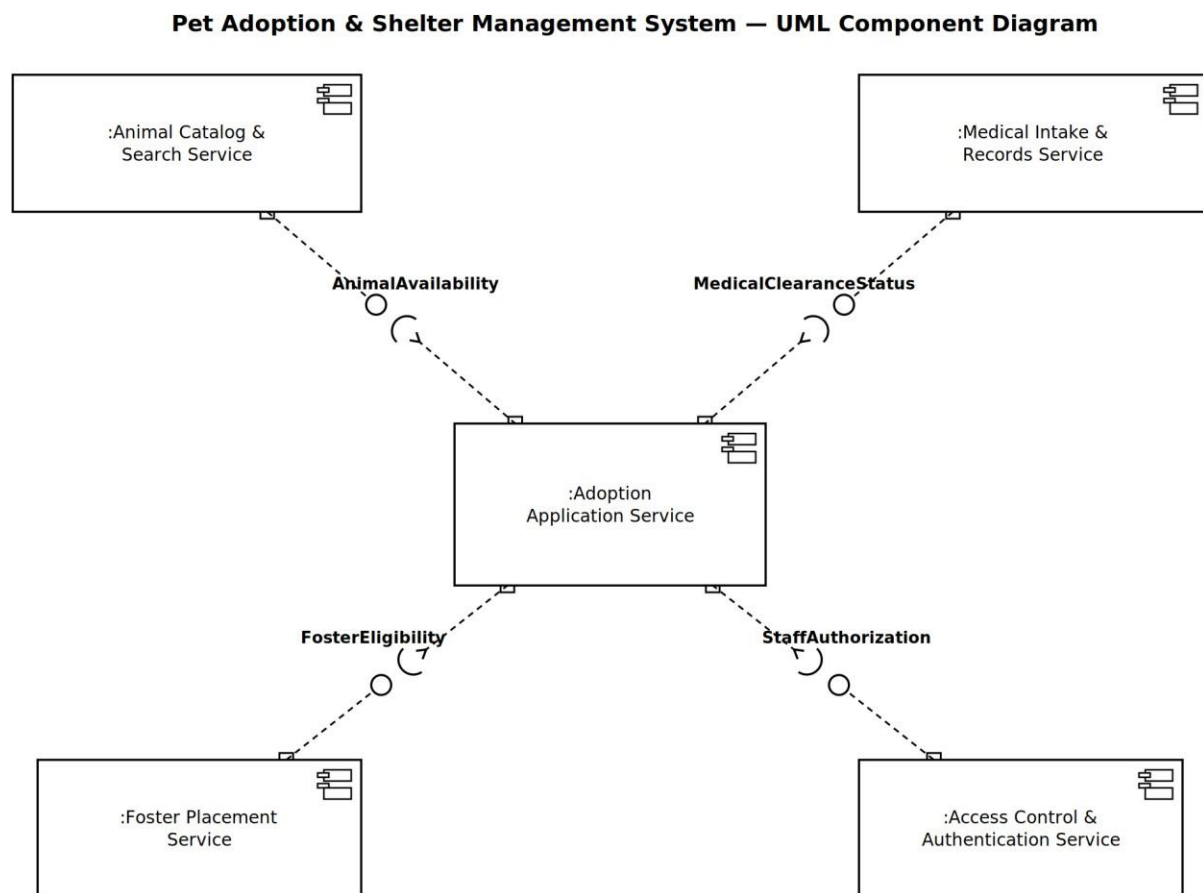
SE-LAB3-COMPONENT DIAGRAM AND ARCHITECTURE

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Component Diagram:



Architectural Justification — Pet Adoption & Shelter Management System

Architecture Selection: We chose the Client-Server Architecture for the Pet Adoption & Shelter Management System.

Reason 1 — A single source of truth is required to prevent double adoption

Multiple actors: adoption applicants browsing and applying, shelter staff reviewing applications, veterinarians logging medical intake, and foster coordinators all act on the same animal records concurrently. FR003 requires that an application for an animal already marked 'Adopted' can never be approved, which is only reliably enforced when one central server holds the authoritative animal and application status. A client-server model keeps every client (staff console, applicant portal) reading and writing through that single controlled data source, ruling out the conflicting-state risk of a fully decentralized design.

Reason 2 — Centralized control simplifies coordinating several distinct workflows

The system spans several linked workflows: medical intake, application review, foster placement that all depend on each other's current state (e.g., approval depends on medical clearance and foster status). A central server can coordinate these dependencies directly, whereas splitting each workflow into independently deployed services would add cross-service consistency overhead disproportionate to the shelter's scale.

Security Advantage

NFR002 restricts medical intake records and applicant personal data to authenticated shelter staff. In a client-server design, the server enforces this centrally: every request for medical or applicant data passes through the Access Control & Authentication Service, which validates the staff session token before the data is released. Client applications (the public applicant portal in particular) never hold direct access to the underlying data store, so an unauthenticated or applicant-role request has no path to sensitive records.

Performance Benefit

NFR001 requires multi-attribute animal search (species, age, breed, compatibility with children) to respond in under 200 ms. Centralizing the animal catalog on the server lets it maintain indexed, up-to-date availability data in one place and serve filtered queries directly, avoiding the network round-trips and data-reconciliation delays that a distributed lookup across independent services would introduce.